

SOLO FULL-STACK PROJECT

v1.0.0-RELEASE

ShopFlow

Enterprise-Grade Distributed Multi-Vendor E-Commerce Platform

Java 17 LTS

Spring Boot 3.3.1

Spring Cloud 2023

Netflix Eureka

Spring Cloud Gateway

PostgreSQL 16

Two-Phase Stock Lock

Stripe Card & UPI

React 18 / TypeScript

Pinterest Editorial UI

ARCHITECTED & ENGINEERED BY

Priyanshu Ghosh

Full-Stack Software Engineer · @PG300604

Architecture Whitepaper
September 2026

1. Executive Summary & Architecture Tenets

ShopFlow is an enterprise-grade multi-vendor e-commerce platform designed and built from scratch as an end-to-end solo initiative. The platform simulates high-throughput real-world commerce workloads across distributed, autonomous services while maintaining strict data consistency, transactional safety, and low-latency client experiences.

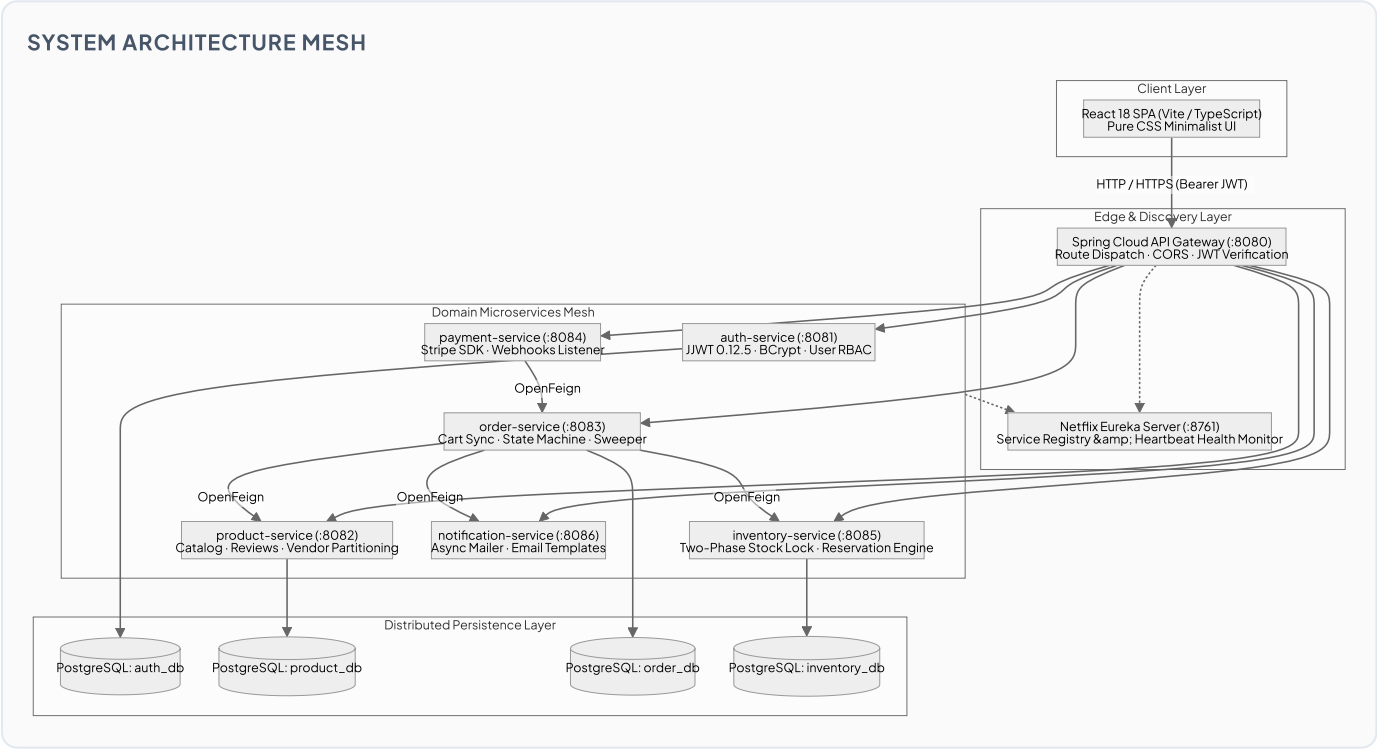
Two-Phase Inventory Lock
Concurrency-safe reservation locks (PENDING → COMMITTED / RELEASED) prevent overselling during checkout spikes.

Stripe Webhooks & Fallback
HMAC-verified webhooks decouple payment authorization from synchronous client requests, backed by client-side fallback order generators.

Multi-Vendor Partitioning
Strict role-based isolation ensures vendors manage independent catalog inventories and query scoped fulfillment pipelines.

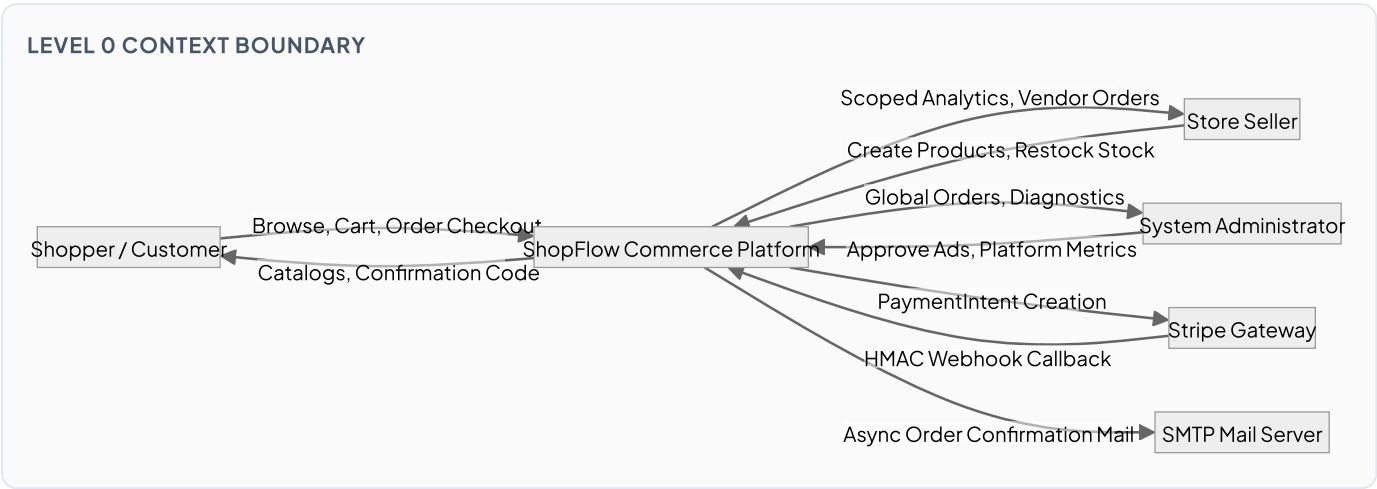
Pinterest Editorial UI
Zero-dependency React 18 + TypeScript frontend with 3-column editorial grid, live swatches, size filters, and full INR (₹) localization.

2. High-Level System Mesh & Edge Topology

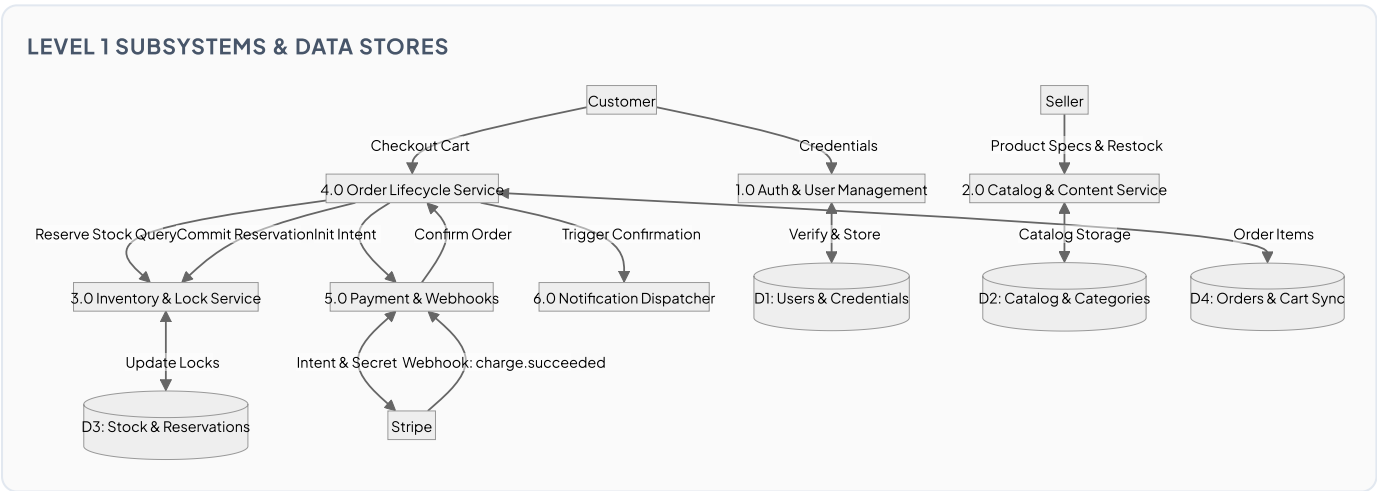


3. Data Flow Diagrams (DFDs)

3.1 DFD Level 0 — Context Diagram

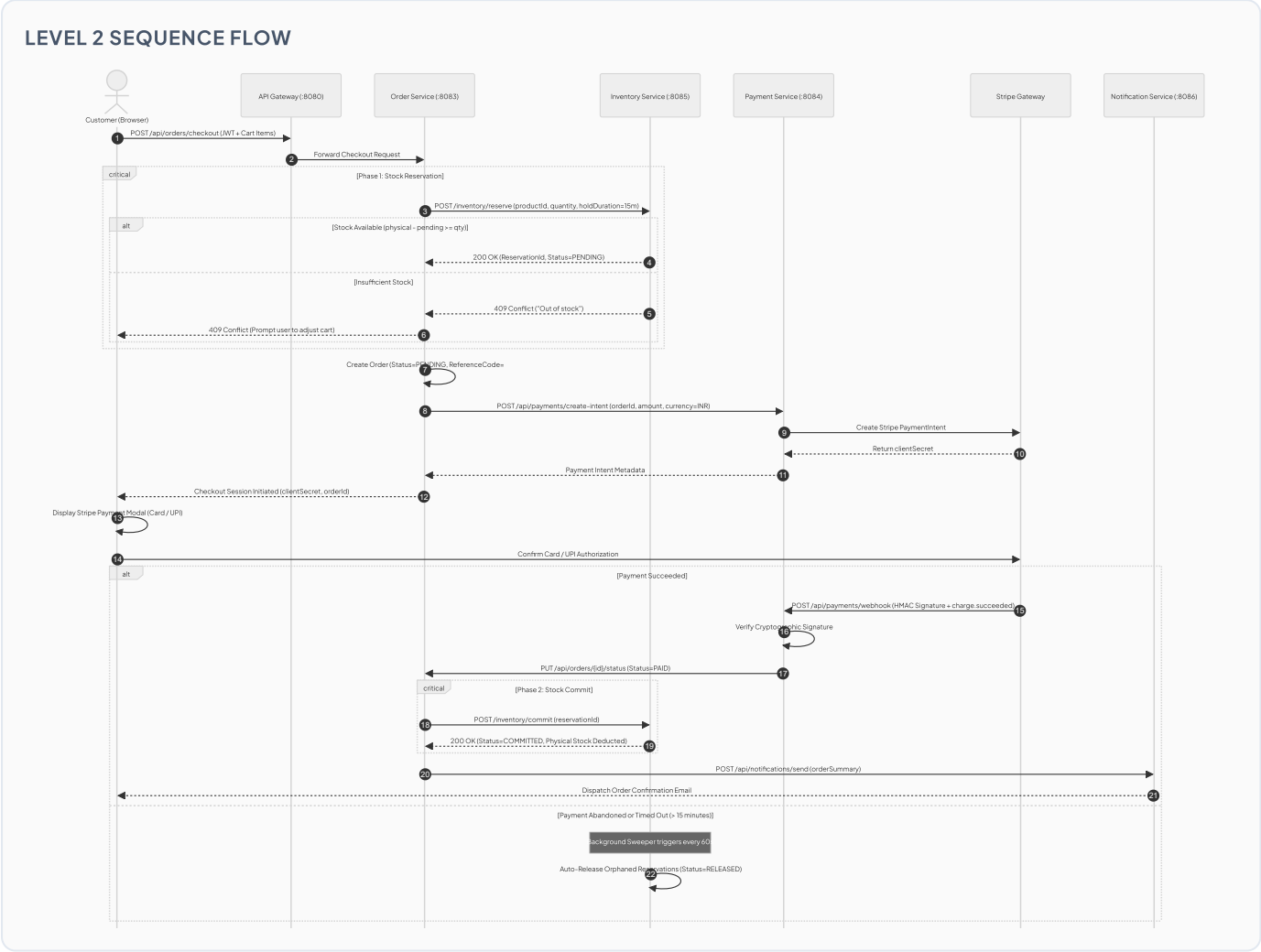


3.2 DFD Level 1 — Core Subsystems Breakdown



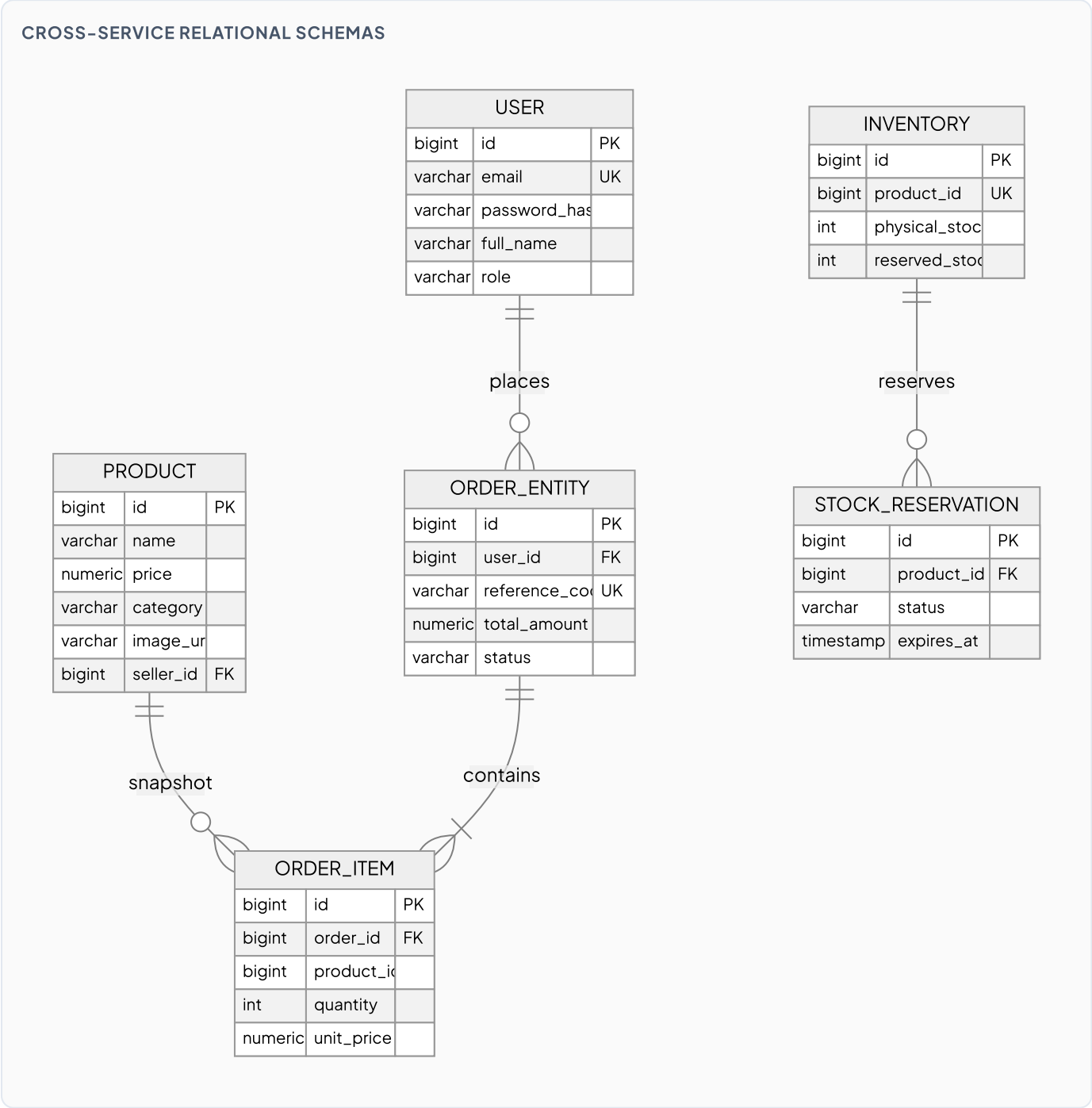
4. End-to-End Checkout & Two-Phase Lock Sequence (DFD Level 2)

The sequence below details the transactional coordination between the shopper, gateway, order service, inventory reservation locks, Stripe webhook dispatch, and final notification commit.

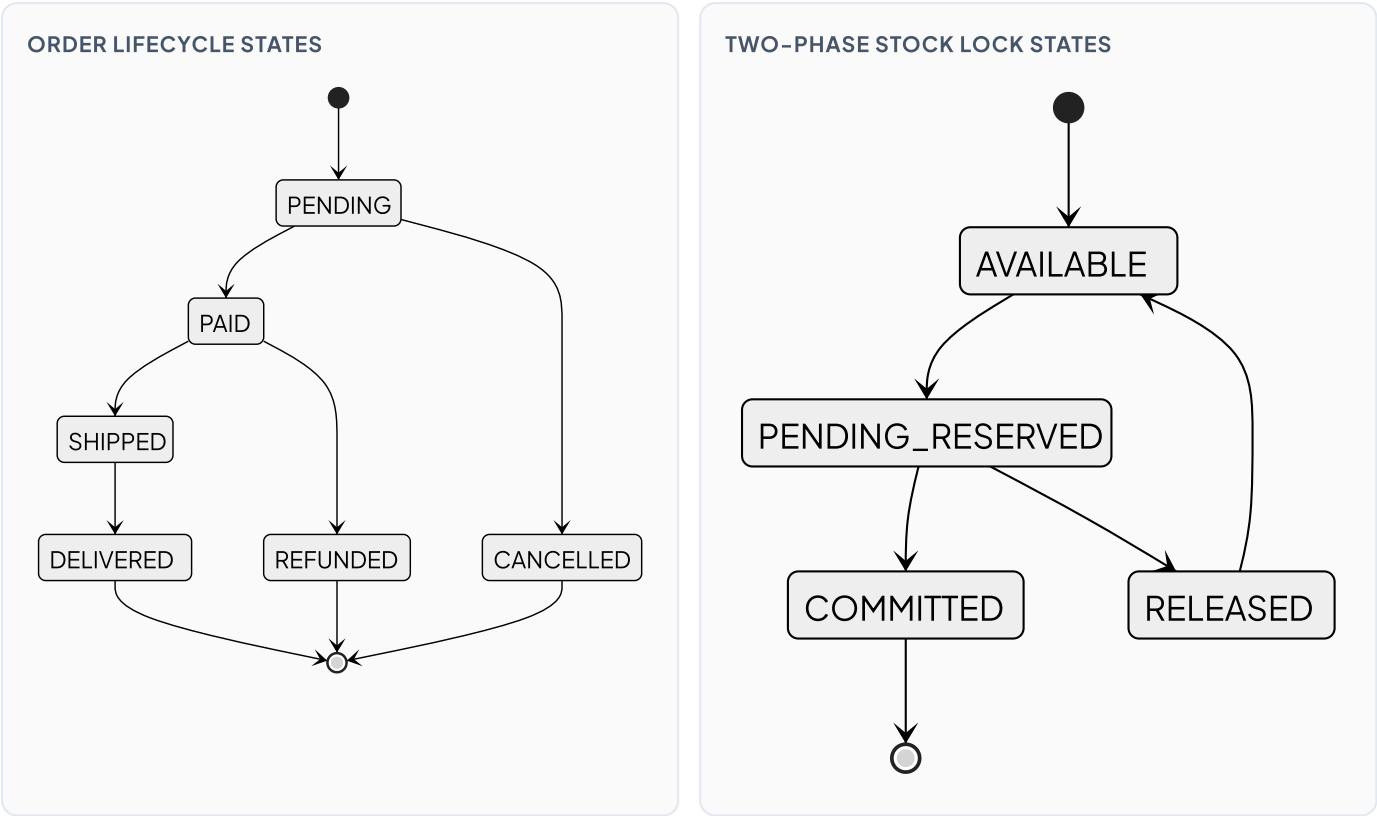


5. Database Entity–Relationship Model (ERD)

Each microservice owns an independent database schema. Cross-service relationships are maintained through immutable identifier snapshotting.



6. Finite State Machines



7. Role-Based Access Control (RBAC) Matrix

Endpoint Pattern	Anonymous	CUSTOMER	SELLER	ADMIN	Enforced Access Scope
POST /api/auth/**	Allowed	Allowed	Allowed	Allowed	Public credentials & JWT issuance
GET /api/products/**	Allowed	Allowed	Allowed	Allowed	Public catalog browsing & details
POST /api/products	Denied	Denied	Allowed	Allowed	Vendor product registration
POST /api/orders/checkout	Denied	Allowed	Allowed	Allowed	Two-phase inventory lock & order creation
GET /api/orders/seller/mine	Denied	Denied	Allowed	Allowed	Orders filtered strictly to vendor products
PUT /api/orders/{id}/admin-status	Denied	Denied	Denied	Allowed	Super-user fulfillment override
POST /api/payments/webhook	Stripe	Stripe	Stripe	Stripe	Cryptographic HMAC signature validation

8. Microservices Registry & Port Registry

Service Name	Port	Base Path	Primary Responsibility	Persistence Layer
api-gateway	8080	/	Edge routing, CORS dispatch, JWT header forwarding	Stateless
eureka-server	8761	/	Service discovery registry, dynamic heartbeat pulse	In-Memory Mesh
auth-service	8081	/api/auth	User account lifecycle, BCrypt, JWT claims issuance	PostgreSQL (auth_db)
product-service	8082	/api/products	Multi-vendor catalog, category indexing, reviews	PostgreSQL (product_db)
order-service	8083	/api/orders	Checkout orchestration, state machine, cart sync	PostgreSQL (order_db)
payment-service	8084	/api/payments	Stripe PaymentIntent generation, webhook listener	Stateless (Feign to Orders)
inventory-service	8085	/inventory	Two-phase reservation locks, 15m auto-sweeper	PostgreSQL (inventory_db)
notification-service	8086	/api/notifications	Async HTML order confirmation via SMTP relay	Stateless
frontend	5173	/	React 18 + Vite storefront with Pinterest grid & INR	LocalStorage / Session

9. Failure Modes & Resilience Patterns

Graceful Fallback Checkout Decoupled modal with resilient client-side order reference generation prevents users from getting stranded during downstream hiccups.	Orphaned Lock Auto-Sweeper Scheduled 60-second background worker automatically sweeps abandoned PENDING stock reservations past 15 minutes.
Spring Security /error Permitted All microservices explicitly permit /error in SecurityFilterChain to ensure domain status exceptions (409, 400) dispatch properly.	Stateless Mesh Load Balancing Netflix Eureka dynamic registry paired with OpenFeign client-side load balancing provides transparent routing without hardcoded IPs.

Engineering Summary: ShopFlow demonstrates complete full-stack mastery across distributed transaction coordination, database schema isolation, zero-framework CSS design, and asynchronous webhook pipelines, architected and built solo by **Priyanshu Ghosh**.